DS604 (Autumn' 2026)

Introduction to Data Management



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“Data Management”

- What is “Data” and What do we manage?
- Different Mathematical representation of the “Data Objects”
- Techniques to Store them on Physical Storage like Hard Disks
- Techniques to
  - Maintain (Add/Modify/Remove)
  - Answer Queries
  - Perform Analytical Tasks

Data and Data Objects (or Data Entities)

- Branch and Customer data objects!

Branch

bid	address	city	state	country	ifsc	micr	swift
B001	Sector-16	Gandhinagar	GJ	IN	IC00165	123456789	ICICINBBXXX
B002	CG Road	Ahmedabad	GJ	IN	IC00136	234567890	ICICINBBYYY
B003	Cannahught Place	New Delhi	ND	IN	IC00011	345678901	ICICINBBZZZ
B004	Nariman Point	Mumbai	MH	IN	IC00004	456789012	ICICINBBAAA
B005	MG Road	Bangalore	KA	IN	IC00056	567890123	ICICINBBBBB
B006	Park Street	Kolkata	WB	IN	IC00099	678901234	ICICINBBCCC
B007	Anna Salai	Chennai	TN	IN	IC00023	789012345	ICICINBBDDD
B008	Jubilee Hills	Hyderabad	TS	IN	IC00088	890123456	ICICINBBEEE
B009	Lavelle Road	Bangalore	KA	IN	IC00077	901234567	ICICINBBFFF
						012345678	ICICINBBGGG

Customer

cno	name	address	email	phone	cclass
C001	Rajesh Kumar	123 Gandhi Nagar, Mumbai	rajesh@example.com	9,876,543,210	Retail
C002	Priya Sharma	456 Rani Road, Delhi	priya@example.com	8,765,432,109	Premium
C003	Amit Patel	789 Subhash Chauraha, Ahmedabad	amit@example.com	7,654,321,098	Retail
C004	Deepak Gupta	101 MG Road, Bangalore	deepak@example.com	6,543,210,987	Student
C005	Meera Reddy	222 Jubilee Hills, Hyderabad	meera@example.com	5,432,109,876	Corporate
C006	Sunita Singh	333 Gomti Nagar, Lucknow	sunita@example.com	4,321,098,765	Retail
C007	Vikram Desai	444 Juhu Beach, Mumbai	vikram@example.com	3,210,987,654	Premium

Data and Data Objects (or Data Entities)

- Transactions' data objects!

Transactions

tid	ts	accountno	description	crdr	amount	balance
T001	2024-06-03	ACC001	Rent	Dr	500	100,000
T002	2024-06-03	ACC002	Groceries	Dr	500	94,500
T003	2024-06-03	ACC003	Medical	Dr	500	189,500
T004	2024-06-03	ACC004	Utilities	Dr	500	98,500
T005	2024-06-03	ACC005	Rent	Dr	500	100,000
T006	2024-06-03	ACC006	Groceries	Dr	500	94,500
T007	2024-06-03	ACC007	Medical	Dr	500	149,500
T008	2024-06-03	ACC008	Utilities	Dr	500	99,500
T009	2024-06-03	ACC009	Rent	Dr	500	99,500
T010	2024-06-03	ACC010	Groceries	Dr	500	99,500
T011	2024-06-03	ACC011	Medical	Dr	500	99,500

“Data Management”

What do we need to manage?

- Store (Add, Modify, Remove)
- Search and answer queries
- Analyse and generate summaries
- Analyse, understand trends, and predict based on trends

Data Storage

Data Manipulation Operations

Analytical Operations



Things that we will learn

- "Data Management" or "Database Management" – principles are almost same
  - Database Collection of Data for a specific objectives
- **Data Models**
  - How do we represent "data object" mathematically
    - ER-Model, Relational, some Non-Relational
  - How do we store data on disks
- **Operations**
  - Relational Algebra
  - SQL
- Principles of executing "queries" (performing operations)

Course Content - more formal

- Database and DBMS concepts
- Relational Theory
- Querying Relational Database using Relational Algebra and SQL
- Design and Implement Relational Databases
- Physical storage of databases and Indexing
- Appreciate how SQL queries are executed in a typical RDBMS
- Modern Generation Databases

Texts

- Not any specific. My notes, though bit concise, should be able to take care.
- You can however refer any of following standard texts-
  - **Fundamentals of Database Systems** by Ramez Elmasri and Shamkant B. Navathe, Pearson Education
  - **Database System Concepts** by Avi Silberschatz, Henry F. Korth, S. Sudarshan, Tata McGraw-Hill

Course Resources

- We will be using Moodle https://moodle.daiict.ac.in/course/view.php?id=245

- **You should be able to self register**

- All course materials, submission of labs and other assignments will be through this page only!

Evaluation for Grade

- Evaluation Strategy:
  - **Exams, labs, and projects that measure how good you have learned the required concepts and learned to apply the concepts in real cases.**
- Marks Distribution
  - Two Mid-Term Exams: 30%
  - Database Project: 20%
  - Lab Evaluation: 20%
  - End Semester examination: 30%

Labs

- DBMS Servers:
  - **PostgreSQL** (10.100.71.21)
  - **MongoDB**
- Lab Exercises (no of labs)
  - Practice Relational Algebra and SQL – 4 weeks
  - Database Design – 2
  - Programming – 2
  - MongoDB
- Group database projects – starts from 5th week. Go up to end of semester; will have certain mile stones (separate document contains more description about projects)

Attendance Policy

- Every student is expected to attend all the lectures.
- You require a minimum of 70%

Lecture Time Table

	MON	TUE	WED	THU	FRI
11:00			DS604		
12:00					
14:00		DS604			
15:00					DS604
16:00					